

FINAL TERM EXAM

Code: CHEM-319 **B.S (Hons) Semester VI** Analytical Chemistry

Time: 2hr **Objective type** Marks: 60

Q1. Choose the correct answer, no over writing and cutting is acceptable. (MCQ'S) (10)

i) Combined action of two complexing reagents in solvent extraction is called?

- a) Batch extraction (b) Continuous extraction
- c) Countercurrent extraction (d) Synergistic extraction

ii) 7. $(C_8H_{17})_3 P=O$ is the general formula of _____

- a) Aliquat 336-S (b) TOPO
- c) TIOA (d) TBP

iii) Flame photometry cannot be used for

- a) Ca (b) Sodium
- c) Cu (d) Li

iv) Laminar flow burner used in Flame photometers is also known as _____

- a) Turbulent burner (b) Premix burner
- c) Total consumption burner (d) Nozzle mix burner

v) The strong base anion exchange resins are usually supplied in _____ form

- a) Hydroxyl (b) Carboxyl
- c) Sulphate (d) Chloride

vi) The exhausted anion exchange column is regenerated by passing a solution of _____

- a) Dil. KOH (b) Conc KOH
- c) Conc. NaOH (d) Dil. NaOH

vii) Gel is basically a colloid in solid form containing _____ % water

- a) 97 (b) 98
- c) 99 (d) 100

viii) Units of zeta potential are _____

- a) Volts (b) Watt
- c) Ohm (d) Joules

ix) In Graphite furnace pyrolytic coating stops the formation of _____ on the graphite tube

- a) Carbide (b) Sulphide c) Nitride (d) Fluoride

x) Incomplete atomization of involatile elements is called _____

- a) Background absorption (b) Memory effect
- c) Physical interference (d) Chemical interference

SUBJECTIVE TYPE

Q 2: Write short answers of the following questions:

(20 marks)

1. Compare Air-acetylene and Nitrous oxide-acetylene flames in AAS?
2. Explain chemical interference with the help of an example?
3. What is the principle of solid phase extraction?
4. What do you mean by the term percent extraction? Explain
5. What are the characteristics of anion exchange resin?
6. Define electroosmotic flow? Write down its mathematical formula?
7. Why it is not suitable to pack the column with dry gel in Gel chromatography?
8. How does ion exchange chromatography help in softening of Hard water?
9. What is the difference between distribution coefficient and distribution ratio in solvent extraction?
10. Why low temperature flame is used for the alkali and alkaline earth metals in FES?

LONG QUESTIONS

Answer the following questions.

(30 marks)

- Q1(a): Explain the construction and working of Graphite furnace with the help of diagram? (5)
- (b) Write down the applications of flame emission spectroscopy? (5)
- Q2(a) Explain different types of factors affecting ion exchange chromatography? (5)
- (b) Explain capillary zone electrophoresis in detail ? (5)
- Q3(a) Explain countercurrent process in solvent extraction? (5)
- (b) Give detail account on gel chromatography? (5)